

Title

Fractional Laser-Primed Photothermal Nanoparticle Therapy Versus Ultrasonic Delivery for Acne Treatment: Clinical and Histological Outcomes

Abstract

Twenty-four Asian patients with moderate-to-severe acne were randomised to either fractional laser followed by nanoparticle therapy or ultrasound-assisted nanoparticle delivery, with both groups subsequently receiving thermal laser treatment. The fractional-laser-primed group demonstrated clinically visible reductions in all lesion categories, including improvement in existing scarring. Histological specimens showed selective destruction of sebaceous follicles with minimal surrounding dermal injury. The ultrasound-assisted delivery group showed comparable clinical responses. Both approaches were well-tolerated with minimal discomfort during treatment and rapid recovery. No participants developed serious adverse effects. Laboratory testing for systemic gold absorption showed negligible serum or urine levels in both groups. Improvement was evident by clinical assessment at follow-up examination. Photographs documented meaningful reduction in acne burden and surface scarring in both arms, with no significant between-group differences. Participants reported high satisfaction with cosmetic outcomes. These findings support the efficacy and safety of both nanoparticle delivery approaches for acne management. The data suggest that laser-based disruption of the stratum corneum does not substantially enhance therapeutic benefit relative to acoustic delivery in this patient population.